

Problem 2.8.2

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Problem Statement

A (6, 1), **B** (8, 2) and **C** (9, 4) are three vertices of a parallelogram $ABCD$.
If E is the midpoint of DC find the area of $\triangle ADE$.

Formula

To Find Area of Triangle

$$\text{Area}(\triangle ADE) = \frac{1}{2} \|(\mathbf{D} - \mathbf{A}) \times (\mathbf{E} - \mathbf{A})\| \quad (1.1)$$

Finding the Point D

$$\mathbf{B} - \mathbf{A} = \mathbf{C} - \mathbf{D} \quad (1.2)$$

$$\mathbf{D} = \mathbf{C} + \mathbf{A} - \mathbf{B} \quad (1.3)$$

$$\mathbf{D} = \begin{pmatrix} 9 \\ 4 \end{pmatrix} + \begin{pmatrix} 6 \\ 1 \end{pmatrix} - \begin{pmatrix} 8 \\ 2 \end{pmatrix} \quad (1.4)$$

$$\mathbf{D} = \begin{pmatrix} 7 \\ 3 \end{pmatrix} \quad (1.5)$$

Finding the midpoint of DC and Area($\triangle ADE$)

$$\mathbf{E} = \frac{\mathbf{D} + \mathbf{C}}{2} = \frac{\begin{pmatrix} 7 \\ 3 \end{pmatrix} + \begin{pmatrix} 9 \\ 4 \end{pmatrix}}{2} = \begin{pmatrix} 8 \\ \frac{7}{2} \end{pmatrix} \quad (1.6)$$

$$\mathbf{D} - \mathbf{A} = \begin{pmatrix} 7 \\ 3 \end{pmatrix} - \begin{pmatrix} 6 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{E} - \mathbf{A} = \begin{pmatrix} 8 \\ \frac{7}{2} \end{pmatrix} - \begin{pmatrix} 6 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ \frac{5}{2} \end{pmatrix} \quad (1.7)$$

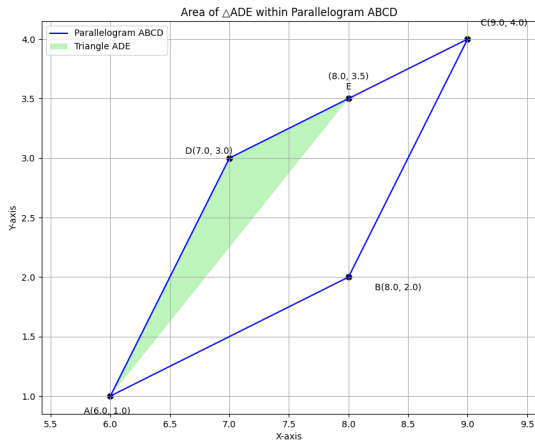
Area($\triangle ADE$)

$$\text{Area}(\triangle ADE) = \frac{1}{2} \|(\mathbf{D} - \mathbf{A}) \times (\mathbf{E} - \mathbf{A})\| \quad (1.8)$$

$$= \frac{1}{2} \left\| \begin{pmatrix} 1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 2 \\ \frac{5}{2} \end{pmatrix} \right\| = \frac{1}{2} \begin{vmatrix} 1 & 2 \\ 2 & \frac{5}{2} \end{vmatrix} = \frac{3}{4} \quad (1.9)$$

Hence the Area of triangle is $\frac{3}{4}$ units.

Plot



Figure

C Code for finding points

```
void calculate_parallelogram_coords(double* out_coords) {  
    // Given vertices  
    double Ax = 6.0, Ay = 1.0;  
    double Bx = 8.0, By = 2.0;  
    double Cx = 9.0, Cy = 4.0;  
  
    double Dx = Ax - Bx + Cx;  
    double Dy = Ay - By + Cy;  
  
    double Ex = (Dx + Cx) / 2.0;  
    double Ey = (Dy + Cy) / 2.0;  
  
    out_coords[0] = Ax; out_coords[1] = Ay;  
    out_coords[2] = Bx; out_coords[3] = By;  
    out_coords[4] = Cx; out_coords[5] = Cy;  
    out_coords[6] = Dx; out_coords[7] = Dy;  
    out_coords[8] = Ex; out_coords[9] = Ey;  
}
```

Calling C Function

```
import ctypes
import numpy as np

def get_all_points():

    # Load the compiled C shared library
    lib = ctypes.CDLL('./coord.so')
    lib.calculate_parallelogram_coords.argtypes = [ctypes.POINTER
        (ctypes.c_double)]

    out_coords_c = double_array_10()

    # Call the C function, passing the C array by reference
    lib.calculate_parallelogram_coords(out_coords_c)

    # Convert the C array result back into a NumPy array
    # Reshape it to 5x2 (5 points, each with x and y)
    all_points = np.array(out_coords_c).reshape(5, 2)
    return all_points
```

Python Code for Plotting

```
#Code by GVV Sharma  
#September 12, 2023  
#Revised July 21, 2024  
#released under GNU GPL  
  
import sys  
import numpy as np  
import matplotlib.pyplot as plt  
sys.path.insert(0, '/workspaces/urban-potato/matgeo/codes/  
    CoordGeo/')  
  
# --- Import from our C Interface Module ---  
from call import get_all_points  
  
from line.funcs import *  
from triangle.funcs import *  
from conics.funcs import circ_gen
```

Python Code for Plotting

```
points=get_all_points()
A, B, C, D, E = points

print(f"Midpoint E coordinates: ({E[0]:.1f}, {E[1]:.1f})")
# Calculate and print the area of triangle ADE
area_ADE = 0.5 * np.abs(A[0]*(D[1] - E[1]) + D[0]*(E[1] - A[1]) +
    E[0]*(A[1] - D[1]))
print(f"The area of Triangle ADE is: {area_ADE}")

# --- Plotting ---
fig, ax = plt.subplots(figsize=(10, 8))

# Draw and fill the shapes
ax.plot(np.vstack([A, B, C, D, A])[:, 0], np.vstack([A, B, C, D,
    A])[:, 1], 'b-', label='Parallelogram ABCD')
ax.fill(np.vstack([A, D, E, A])[:, 0], np.vstack([A, D, E, A])[:,
    1], 'lightgreen', alpha=0.6, label='Triangle ADE')
```

Python Code for Plotting

```
# Draw the vertices
```

```
ax.scatter(points[:, 0], points[:, 1], color='black', s=40)
```

```
# Add annotations with specific offsets
```

```
ax.annotate(f'A({A[0]:.1f}, {A[1]:.1f})', xy=A, xytext=(-30, -20),  
           textcoords='offset points')
```

```
ax.annotate(f'B({B[0]:.1f}, {B[1]:.1f})', xy=B, xytext=(30, -15),  
           textcoords='offset points')
```

```
ax.annotate(f'C({C[0]:.1f}, {C[1]:.1f})', xy=C, xytext=(15, 15),  
           textcoords='offset points')
```

```
ax.annotate(f'D({D[0]:.1f}, {D[1]:.1f})', xy=D, xytext=(-50, 5),  
           textcoords='offset points')
```

```
ax.annotate(f'({E[0]:.1f}, {E[1]:.1f})\nE', xy=E, xytext=(0, 8),  
           textcoords='offset points', ha='center', va='bottom')
```

Python Code for Plotting

```
# --- Final Formatting ---  
ax.set_xlabel('X-axis')  
ax.set_ylabel('Y-axis')  
ax.set_title('Area of Triangle ADE within Parallelogram ABCD')  
ax.grid(True)  
ax.axis('equal')  
ax.legend()  
plt.show()  
plt.savefig('../figs/fig1.png')
```